Regime Switching Model of ALSFRS-R Attrition and Respiratory Reserve Delineates Homeostatic Tipping Boundaries in Amyotrophic Lateral Sclerosis

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Abstract

Amyotrophic lateral sclerosis (ALS) exhibits substantial heterogeneity in functional deterioration and respiratory decline, complicating the interpretation of longitudinal ALS Functional Rating Scale Revised (ALSFRS-R) and vital capacity measurements. Although established prognostic approaches characterize disease severity and survival, the mathematical conditions under which coupled functional and respiratory deterioration crosses a defined compensation boundary remain insufficiently examined. This study analysed whether a literature-parameterized nonlinear model could characterize such transitions using routinely collected clinical measurements without assuming patient level calibration. We developed a four-variable dynamical system comprising normalized functional reserve (F), respiratory reserve (R), latent disease momentum (M), and homeostatic reserve (H). A weighted homeostatic margin, $\Psi(t) = w_F F(t) + w_R R(t) + w_H H(t) - \theta_0 - \theta_M M(t)$, classified simulated trajectories into preserved, compensated, transitional, and decompensated regimes. Mathematical evaluation included boundedness analysis, frozen observable quasi equilibria, Jacobian eigenvalues, phase plane analysis, and continuation of a 12-month functional respiratory compensation boundary. Computational robustness was examined through five clinical scenarios, 500 Latin-hypercube parameter sets with partial rank correlation analysis, 500 stochastic Euler Maruyama trajectories, and independent numerical solver checks. Synthetic model recovery comparisons and an exploratory sequential monitoring analysis were conducted. The reference simulation reproduced early ALSFRS-R and forced vital capacity (FVC) declines of 1.04 points/month and 2.27 percentage points/month, respectively. The homeostatic margin crossed zero at 13.70 months, with simulated ALSFRS-R/FVC values of 25.57/59.12% at month 12 and 17.56/41.60% at month 18. Across 442 baseline functional respiratory combinations, 323 (73.1%) produced negative 12-month margins, 104 (23.5%) occupied the transitional interval, and 15 (3.4%) retained margins ≥0.10. At an initial ALSFRS-R of 40, the model shows minimum FVC required to maintain a nonnegative 12-month margin was 80% predicted. Respiratory vulnerable, accelerated functional decline, and low combined reserve scenarios crossed the boundary at 9.65, 10.00, and 6.75 months, respectively, whereas the slow progression scenario remained above it throughout 18 months. Frozen observable latent equilibria were locally stable, with dominant real Jacobian eigenvalues ranging from -0.107 to -0.072. Global parameter sampling yielded a median terminal margin of -0.151 (2.5^{th}-97.5^{th} percentile: -0.309 to -0.002), with 97.8% of parameter sets the boundary by 18 months. Fixed parameter stochastic perturbations produced a median first passage time of 13.65 months (12.27-15.15). Respiratory attrition (PRCC = -0.855) and momentum damping by homeostatic reserve (PRCC = 0.852) were prominent dynamic determinants of terminal margin. Coupled functional and respiratory attrition produced a reproducible, parameter dependent compensation boundary within a nonlinear ALS model, while latent state stability and first passage analyses distinguished boundary crossing from conventional equilibrium instability. The output support a testable mathematical representation of functional respiratory deterioration and identify structural parameters requiring empirical estimation. These results arise entirely from literature oriented assumptions and simulations, not observed patient transitions. Longitudinal calibration, parameter identifiability assessment, and external validation are necessary before the proposed boundaries or monitoring decisions can inform clinical care.

Keywords: *ALSFRS-R, Amyotrophic Lateral Sclerosis, Nonlinear Dynamical Systems, Regime Switching Classification, Respiratory Function*

1. Introduction

Amyotrophic lateral sclerosis (ALS) is a progressive motor neuron disorder characterized by heterogeneous rates of functional decline and respiratory impairment. The timing and relative contribution of these changes vary considerably between individuals, complicating longitudinal assessment and clinical planning (Westeneng et al., 2018 and Feldman et al., 2022). The revised ALS Functional Rating Scale (ALSFRS-R) summarizes functional limitations across bulbar, fine motor, gross motor and respiratory domains on a 0-48 scale (Cedarbaum et al., 1999). Forced vital capacity (FVC) and slow vital capacity (SVC) provide complementary measurements of respiratory function, although neither measure alone fully characterizes the preceding course of functional deterioration (Andrews et al., 2018 and Pinto and de Carvalho, 2019). Longitudinal clinical trial data provide empirical context for the rates of change represented by these measures. In the original Pooled Resource Open-Access ALS Clinical Trials (PRO-ACT) analysis, mean declines were 1.02 ALSFRS-R points per month and 2.24 percentage points of predicted vital capacity per month (Atassi et al., 2014). In a separate analysis of placebo treated trial participants, decrease in SVC was associated with respiratory insufficiency and related outcomes (Andrews et al., 2018). Pinto and de Carvalho (2019) also reported a strong correlation between FVC and SVC and found that their longitudinal decreases were associated with functional deterioration. These findings support joint consideration of functional and respiratory trajectories, but cohort level rates should not be interpreted as individual specific disease courses or as direct estimates of coefficients in a mechanistic model.

Existing approaches address clinically important but different questions. King's clinical staging summarizes disease milestones (Balendra et al., 2019), whereas the MiToS system classifies loss of independence across major functional domains (Chiò et al., 2015). The externally validated ENCALS model combines clinical predictors to estimate individualized survival outcomes (Westeneng et al., 2018). These methods provide useful staging and prognostic information. Their principal reported outputs differ from the specific mathematical quantity considered here, the first passage of a coupled functional respiratory trajectory across a prespecified model compensation boundary. A nonlinear state space model offers a way to investigate this quantity explicitly. In the present study, normalized ALSFRS-R and vital capacity are represented by observable functional and respiratory reserves, $F(t)$ and $R(t)$, and coupled to two unobserved constructs such as disease momentum, $M(t)$, and homeostatic reserve, $H(t)$. These latent quantities are mathematical hypotheses. A weighted margin, $\Psi(t)$, refers an operational boundary and permits classification of simulated trajectories into four ordered reserve regimes. Crossing $\Psi = 0$ is a model first passage event, not evidence of a biological bifurcation, an established ALS stage, or a validated clinical threshold. The term regime denotes classification by the margin unless an independently specified stochastic transition process is implemented and evaluated. The analytical contribution is to examine how this boundary depends on initial functional respiratory status, structural parameters and uncertainty. Frozen observable quasi equilibria and Jacobian eigenvalues assess local behavior of the latent subsystem, numerical continuation describes the compensation surface, and scenario analyses, latin-hypercube sampling, partial rank correlations and stochastic perturbations examine the sensitivity of first passage timing. Synthetic model recovery and exploratory monitoring decisions are treated as methodological demonstrations. Although these techniques also appear in quantitative risk and decision analysis, their use here is justified by the mathematical questions posed.

The aim of this literature-parameterized proof-of-concept is to determine whether a parsimonious coupled model, oriented toward published ALSFRS-R and vital capacity decline rates, can produce reproducible and falsifiable functional respiratory compensation boundaries. No patient level observations were fitted, and the model's latent parameters, regime cutoffs and decision costs are provisional. Accordingly,

the study does not estimate individual prognosis or recommend treatment or monitoring thresholds. Its intended contribution is to define explicit mathematical hypotheses, identify parameters requiring estimation and establish a reproducible basis for subsequent longitudinal calibration, identifiability analysis and external validation using appropriately accessed ALS cohorts.

2. Methods

2.1 Study Design

We adopted a literature computational proof-of-concept model, distinguishing observed clinical anchors from provisional model coefficients. Published ALS functional and respiratory evidence informed variable selection and approximate decline rates (Cedarbaum et al., 1999 and Andrews et al., 2018). The mathematical model draws on nonlinear dynamical systems analysis (Strogatz, 2018), while uncertainty experiments are informed by global sensitivity methods (Marino et al., 2008). These methodological references support the analytical techniques. We conducted a deterministic and stochastic computational investigation of coupled functional and respiratory attrition in amyotrophic lateral sclerosis (ALS). Observable definitions and approximate decrease rates were drawn from published ALS research, while latent state kinetics, compensation margin coefficients, and monitoring costs were specified as provisional structural assumptions. The original PRO-ACT analysis supplied empirical orientation for early ALSFRS-R and vital capacity decline (Atassi et al., 2014), no PRO-ACT records or individual-level outcomes were obtained or fitted in this study. Consequently, all numerical results are conditional model outputs, not patient specific predictions, observed transition rates, or validated clinical thresholds. The architecture follows the conceptual sequence from observable clinical anchors to continuous trajectories, margin regime labels, mathematical analysis, and exploratory decisions.

2.2 Observable Literature Anchors and Analytical Roles

The ALS Functional Rating Scale-Revised (ALSFRS-R) ranges from 0 to 48 and includes respiratory items (Cedarbaum et al., 1999). We denote normalized functional status $F(t) = ALSFRS - R(t)/48$ and normalized respiratory status $R(t) = VC(t)/100$, where VC denotes percentage predicted forced vital capacity (FVC) in the primary simulation. Slow vital capacity (SVC) was considered a related, but not interchangeable, clinical measurement because FVC and SVC have been examined comparatively in ALS (Pinto and de Carvalho, 2019). Reference baseline values of ALSFRS-R 40 and FVC 90.5% predicted were illustrative choices. Target early slopes were informed by mean PRO-ACT changes of 1.02 ALSFRS-R points and 2.24 vital capacity percentage points per month (Atassi et al., 2014). The placebo trial analysis of SVC decline provides additional respiratory context, but its SVC rates were not treated as FVC specific calibration observations (Andrews et al., 2018).

Table 1. Literature anchors and analytical roles.

Quantity	Value / definition	Status and role
ALSFRS-R	0-48	Validated clinical scale (Cedarbaum et al., 1999)
PRO-ACT ALSFRS-R	1.02 points/month	Published mean, orientation only (Atassi et al., 2014)
PRO-ACT vital capacity	2.24 percentage points/month	Published mean, orientation only (Atassi et al., 2014)
SVC	~2.7 percentage points/month	Separate trial context benchmark (Andrews et al., 2018)
Reference baseline	ALSFRS-R 40, FVC 90.5%	Illustrative model inputs
Reference lines	ALSFRS-R 38, FVC 80%	Visual orientation only

2.3 Nonlinear State Model

The coupled nonlinear ordinary differential equations were modeled using standard state space reasoning (Strogatz, 2018). The equations below are original, provisional structural assumptions. The state vector $x(t) = [F(t), R(t), M(t), H(t)]^{\mathrm{T}}$ comprises normalized functional and respiratory quantities and two latent mathematical constructs: disease momentum M and homeostatic reserve H. Time t is measured in months and all four states are dimensionless. The following coupled ordinary differential equations:

$$\frac{dF}{dt} = -k_F\, MF - k_{FR}(1-R)F$$

$$\frac{dR}{dt} = -k_R\, MR - k_{RF}(1-F)R$$

$$\frac{dM}{dt} = k_{M0}(1-M) + k_{MF}(1-F)(1-M) + k_{MR}(1-R)(1-M) + k_{MM}(1-M) - k_{MH}\, HM$$

$$\frac{dH}{dt} = k_{HF}\, F(1-H) + k_{HR}\, R(1-H) - k_{HM} MH - k_{Hd}\, H$$

The boundary sign argument follows the forward invariance reasoning used for nonlinear differential systems (Strogatz, 2018). All kinetic coefficients are nonnegative and have unit month^{-1}. At F =0 and $R = 0$ the respective derivatives vanish. At $M = 0$ and $H = 0$ their derivatives are nonnegative. At $M = 1$ and $H = 1$ they are nonpositive. Thus, the vector field is inward pointing or tangent on the boundary of $[0,1]^4$, establishing forward invariance for normalized initial conditions and nonnegative coefficients. The model assumes monotonic F and R in its deterministic implementation and cannot represent clinical improvement or treatment recovery. The model compensation margin was calculated as

$$\Psi(t) = w_F\, F(t) + w_R\, R(t) + w_H\, H(t) - \theta_0 - \theta_M\, M(t)$$

The use of a time varying margin and first passage surface is conceptually related to critical transition analysis (Scheffer et al., 2009), but the chosen margin and its thresholds are model only and do not show a biological tipping point bifurcation. The reference weights were $w_F = 0.34, w_R = 0.34, w_H = 0.32, \theta_0 = 0.34$ and $\theta_M = 0.28$. These are structural choices. Regime labels were assigned deterministically, preserved ($\Psi > 0.25$), compensated ($0.10 < \Psi \leq 0.25$), transitional ($0 < \Psi \leq 0.10$), and decompensated ($\Psi \leq 0$). The first passage time was defined as $\tau * = inf\{t \geq 0 : \Psi(t) \leq 0\}$. When no crossing occurred within 18 months it was recorded as not observed in the simulation horizon. A crossing of $\Psi = 0$ is a model event and is not evidence of a biological bifurcation or clinical deterioration event. The exponential index is an illustrative mathematical score. A separate, model hazard index was evaluated as $h(t) = h0 \exp[b_M\, M + bF(1-F) + b_R(1-R) - b_H\, H]$, with $h_0 = 0.015, b_M = 1.8, b_F = 1.2, b_R = 1.5$, and $b_H = 1.0$. Reported multipliers are $\frac{h(t)}{h(0)}$. No event time likelihood was fitted, so this quantity must not be interpreted as an empirical hazard ratio or patient event rate.

Table 2. Reference structural coefficients used by the supplied simulation code. All values are provisional assumptions and not patient fitted.

Parameter	Reference value
k_F	0.08
k_{FR}	0.018
k_R	0.075
k_{RF}	0.015

k_{M0}	0.02
k_{MF}	0.08
k_{MR}	0.1
k_{MM}	0.09
k_{MH}	0.1
k_{HF}	0.035
k_{HR}	0.045
k_{HM}	0.1
k_{Hd}	0.012
θ_0	0.34
θ_M	0.28
w_F	0.34
w_R	0.34
w_H	0.32
h_0	0.015
b_M	1.8
b_F	1.2
b_R	1.5
b_H	1.0
w_H	0.32
h_0	0.015

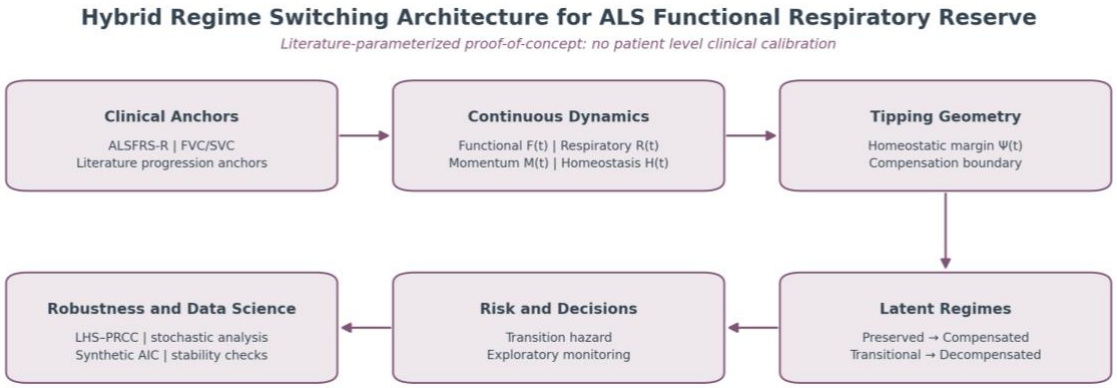


Figure 1. Conceptual architecture of the literature-parameterized ALS functional respiratory compensation model

2.4 Quasi Equilibrium, Local Stability and Boundary Continuation

Local stability was assessed using Jacobian eigenvalues of the frozen observable latent subsystem, following standard nonlinear stability analysis (Strogatz, 2018). To characterize latent state behavior without implying that ALS reaches a stationary disease state, F and R were held fixed at three illustrative clinical anchors: ALSFRS-R/FVC $42/95\%, 38/80\%$, and $30/60\%$. The two-dimensional M-H subsystem was solved for quasi equilibria and its Jacobian was approximated numerically. The real parts of its eigenvalues were used to assess local asymptotic stability conditional on frozen observable states. A phase plane stream field was constructed at ALSFRS-R 38 and FVC 80%. This analysis does not establish stability of the evolving full four-dimensional system. Separately, baseline ALSFRS-R and FVC were varied to evaluate Ψ at month 12 and to identify the minimum grid FVC producing $\Psi(12) \geq 0$. This compensation boundary is conditional on fixed latent initial states and chosen coefficients.

2.5 Numerical Solution and Scenarios

The numerical integration employed an embedded Runge-Kutta method (Dormand & Prince, 1980), solver comparison and step size checks follow established numerical ODE practice (Hairer et al., 1993). The

reference initial condition was $x(0) = (40/48, 0.905, 0.22, 0.78)$, with an 18-month horizon and output spacing of 0.05 months. The ODEs were solved using $scipy.integrate.solve_ivp$ with RK45 (relative tolerance 10^{-8}, absolute tolerance 10^{-10}). Numerical agreement was assessed against DOP853 using maximum step of 1.0, 0.5, 0.25, and 0.10 months. A 17×26 baseline grid comprised ALSFRS-R values 28-44 and FVC values 50-100% predicted. Five scenarios were specified a priori by altering initial states and selected rates, scenario choices were not inferred from patient subgroups. The code records first passage time on the 0.05-month output grid, so reported crossing times have corresponding grid resolution.

Table 3. Scenario Initial Values from the Reference Simulation

Scenario	Initial F, R, M, H
Slow progression	0.875, 0.95, 0.16, 0.84
PRO-ACT	0.8333333333333334, 0.905, 0.22, 0.78
Respiratory vulnerable	0.8333333333333334, 0.8, 0.28, 0.7
Fast functional progression	0.7916666666666666, 0.88, 0.3, 0.7
Low combined reserve	0.7083333333333334, 0.7, 0.32, 0.62

2.6 Structural sensitivity and stochastic perturbations

Latin-hypercube sampling and partial rank correlation coefficients (PRCCs) were selected in accordance with global uncertainty and sensitivity analysis practice (Marino et al., 2008). The stochastic update used Euler-Maruyama discretization (Kloeden & Platen, 1992). Global structural uncertainty was assessed using 500 Latin-hypercube draws from prespecified uniform coefficient intervals. The sampled ranges are sensitivity assumptions. Outputs included terminal Ψ, first passage time, and terminal state variables. Partial rank correlation coefficients (PRCCs) were computed by ranking each input and output, regressing each ranked variable on the other ranked inputs, and correlating the residuals. The implementation also calculates a one-sided +1% finite difference relative sensitivity of terminal margin, normalized by the absolute baseline terminal margin. This should not be presented as a conventional logarithmic elasticity when the output is negative. The fixed parameter stochastic experiment achieved 500 Euler-Maruyama paths with step $\Delta t = 0.05$ months, independent Gaussian increments, diffusion amplitudes (0.005, 0.006, 0.015, 0.012) for (F, R, M, H), and projection to [0,1] after each update. This projection is a numerical modeling choice. The diffusion terms were not fitted to biological variation or measurement error.

$$X_{n+1} = clip_{(0,1)}[X_n + f(X_n; \theta)\Delta t + diag(\sigma)\sqrt{\Delta t}\,\varepsilon_n], \quad \varepsilon_n \sim N(0, I).$$

2.7 Synthetic Model Recovery Comparison

The internal comparison used an information criterion approach motivated by Akaike (1974), applied here to standardized synthetic residuals. A separate internal benchmark used monthly pseudo-observations over months 0-12, achived from the reference structural trajectory with independent Gaussian errors (standard deviations 0.45 ALSFRS-R points and 1.8 FVC percentage points). Four candidate curve specifications were compared such as the structural ODE model, a coupled exponential model, independent linear trends, and a shared-inflection logistic model. Residuals were standardized by the stated noise scales and the implemented criterion was $AIC* = n\log\left(\frac{RSS}{n}\right) + 2k$, with $n = 26$ stacked observations and k equal to the number of fitted coefficients assigned to each candidate. Because data were simulated from the structural model, these values quantify internal recovery under one synthetic experiment, not clinical discrimination or external predictive performance. A code-level limitation is that θ_0, counted among the four fitted structural

parameters, does not affect the simulated ALSFRS-R/FVC observations. The reported structural $AIC*$ should be considered provisional until the effective parameter count and comparison are corrected. We retain this benchmark as exploratory and do not use it to support clinical superiority.

2.8 Exploratory Decision Surface

Sequential monitoring can be studied using stochastic dynamic programming (Puterman, 1994), but the present implementation is a fixed deterministic threshold rule. For baseline ALSFRS-R values 30-44 and FVC values 50-100% in increments of two, the reference ODE system was propagated three months using latent initial conditions $M(0) = 0.22$ and $H(0) = 0.78$. The terminal margin $\Psi(3)$ was transformed into an illustrative decompensation score $s = \frac{1}{[1+\exp(10\Psi(3))]}$. This is a deterministic threshold rule with arbitrary costs ($waiting\ cost = s, monitoring\ cost = 0.20$). The output illustrates how later patient model might be used to examine monitoring schedules, subject to clinician actions and utilities.

3. Results

3.1 Reference Dynamics and Literature Initial Condition

The reference initial condition was ALSFRS-R 40, FVC 90.5% predicted, disease momentum M = 0.22, and homeostatic reserve H = 0.78. Over months 0-6, the model yielded mean slopes of -1.037 ALSFRS-R points/month and -2.268 FVC percentage points/month, compared with the published PRO-ACT orientation values of -1.02 and -2.24, respectively (Atassi et al., 2014). This proximity was intentional and is not independent external validation. At month 6, simulated ALSFRS-R was 33.78 and FVC 76.89%. At month 12, these values were 25.57 and 59.12%. The first sampled $\Psi \leq 0$ occurred at 13.70 months. By month 18, ALSFRS-R was 17.56, FVC 41.60%, and $\Psi = -0.138742$. The model hazard index rose to 12.95 times its initial value.

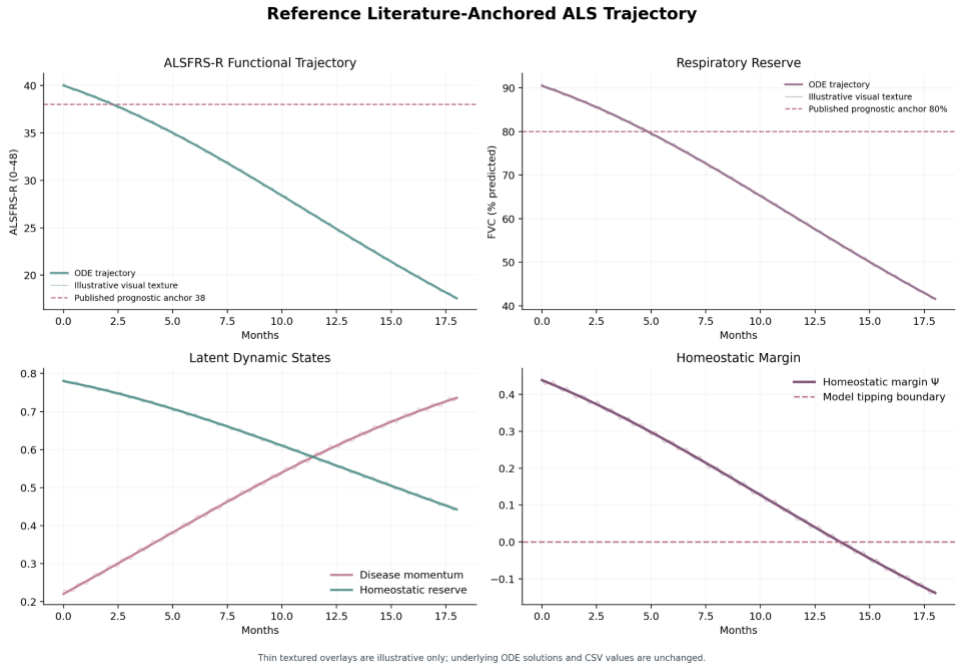


Figure 2. Reference functional and respiratory trajectories, latent momentum and reserve, and model homeostatic margin. Thin textured overlays are plotted ODE numerical solution result.

3.2 Twelve-Month Surface Output and Continuation

The baseline grid comprised 442 combinations (17 ALSFRS-R values × 26 FVC values). At month 12, 323 cells (73.1%) had $\Psi < 0$, 104 (23.5%) had $0 \leq \Psi < 0.10$, and 15 (3.4%) had $\Psi \geq 0.10$. The contour $\Psi(12) = 0$ is conditional on the common latent initial states and provisional coefficients. Continuation on the discrete baseline grid gave the minimum FVC compatible with a nonnegative month-12 margin: 99% at ALSFRS-R 30, 83% at 38, 80% at 40, and 73% at 44. The staircase is a grid resolution artifact. The FVC 80% line is a visual reference only.

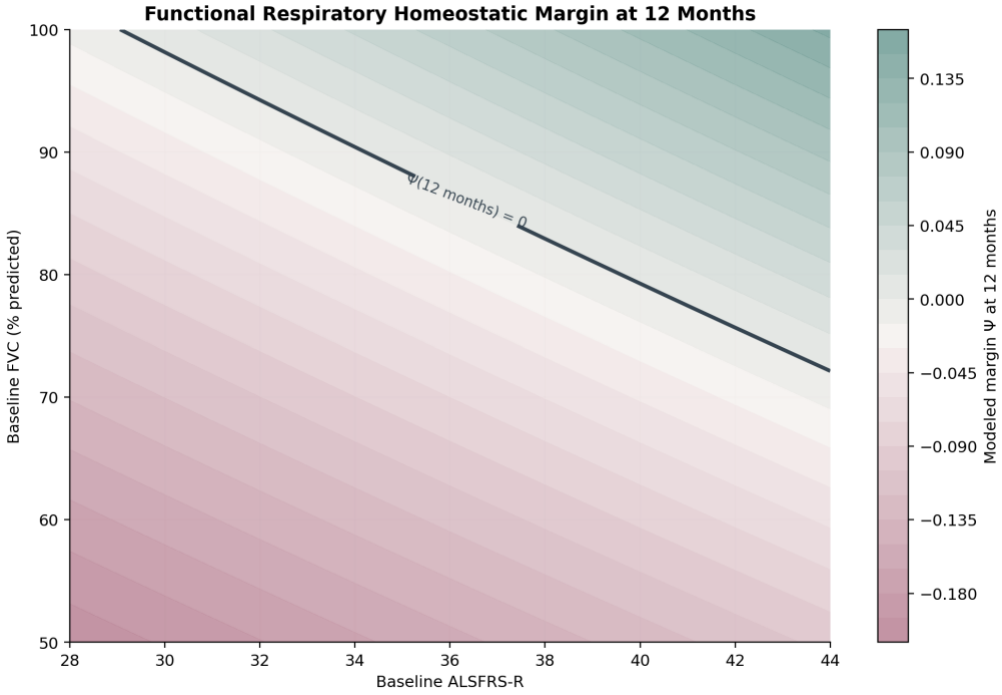


Figure 3. Modeled 12-month compensation margin over baseline ALSFRS-R and FVC. Dark contour: $\Psi(12) = 0$.

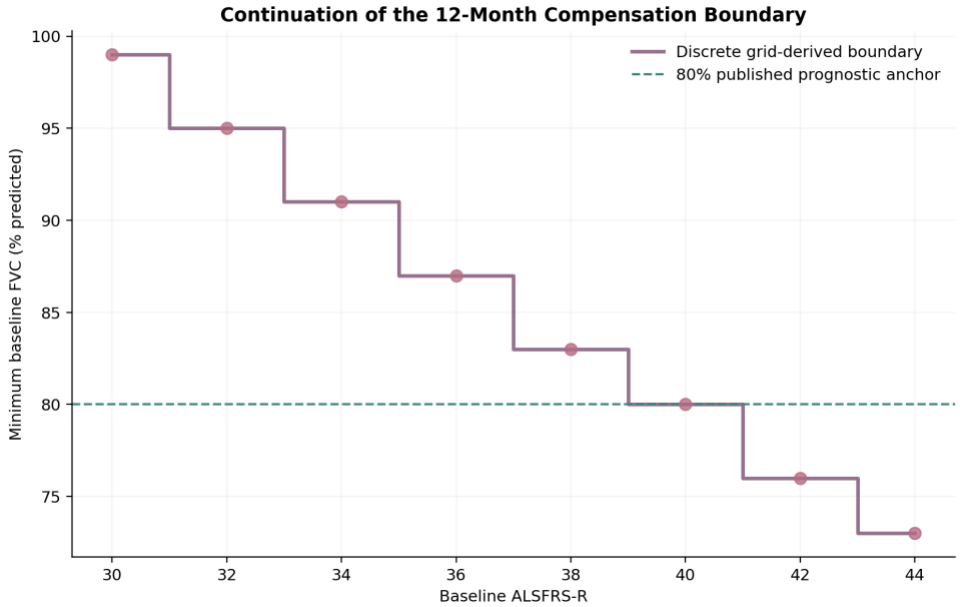


Figure 4. Discrete continuation of minimum baseline FVC yielding $\Psi(12) \geq 0$. Stair steps reflect the sampling grid, illustrating dashed 80% line.

3.3 Prespecified scenario stress tests

Five structural scenarios altered baseline reserves and/or selected rate coefficients. The slow progression scenario had no observed crossing within 18 months and retained a positive terminal margin ($\Psi = 0.041$). The reference crossed at 13.70 months, whereas respiratory vulnerability, faster functional loss, and low combined reserve shifted the first passage time to 9.65, 10.00, and 6.75 months, respectively. The contrasting 12-month ALSFRS-R/FVC pairs under respiratory vulnerability (22.53/39.63%) and fast functional progression (18.16/51.58%) illustrate different simulated routes toward negative compensation margins.

Table 4. Stress Test Scenario Result

Scenario	Crossing (mo)	ALSFRS-R 12 mo	FVC 12 mo (%)	Ψ 18 mo	Index multiplier
Slow progression	Not by 18	32.82	74.56	0.041	8.07
PRO-ACT-reference	13.70	25.57	59.12	-0.139	12.95
Respiratory vulnerable	9.65	22.53	39.63	-0.276	16.66
Fast functional progression	10.00	18.16	51.58	-0.262	15.82
Low combined reserve	6.75	16.15	33.84	-0.344	13.76

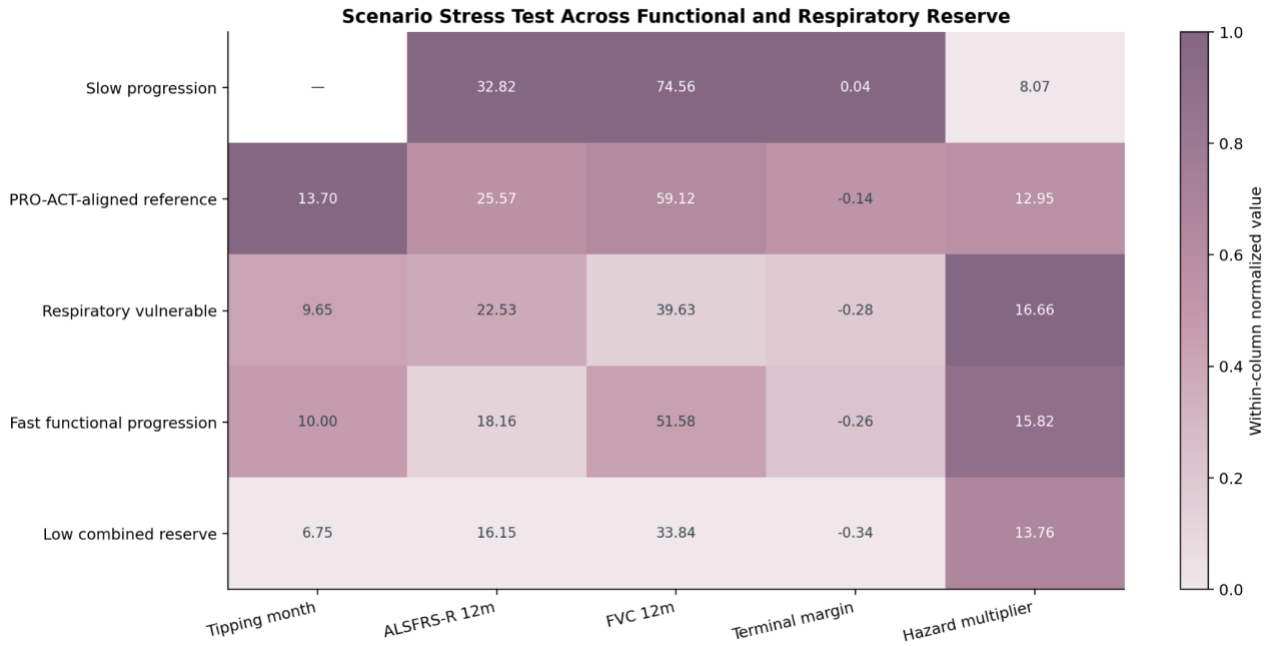


Figure 5. Heatmap showing scenario level crossing times, month-12 outcomes, terminal margin and model hazard index multiplier.

3.4 Structural Sensitivity and Parameter Uncertainty

Across 500 Latin-hypercube parameter draws, the median month-18 margin was -0.1507 (2.5^{th}-97.5^{th} simulation percentiles -0.3094 to -0.0016), 97.8% of parameter sets crossed $\Psi = 0$ by month 18. These

intervals describe chosen parameter ranges. The largest absolute partial rank correlations with terminal margin were θ_M (-0.888), θ_0 (-0.856), k_R (-0.855), k_{MH} (+0.852), k_F (-0.813), and k_{HM} (-0.792). Sensitivity to θ_M and θ_0 underscores dependence on the arbitrarily specified boundary, sensitivity to rate coefficients shows which dynamical assumptions warrant future estimation.

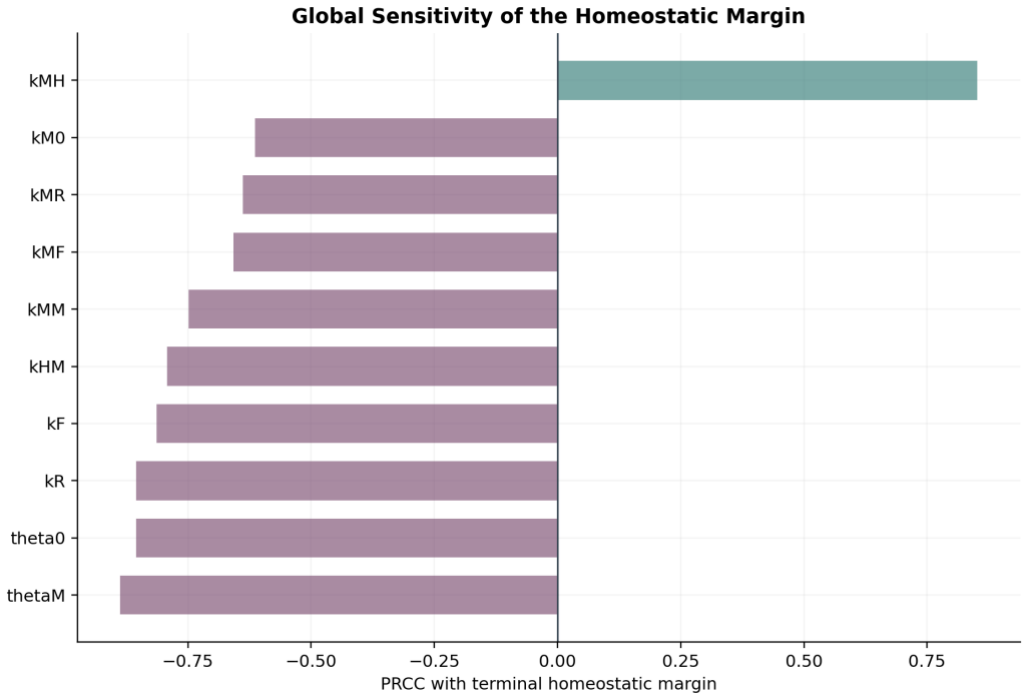


Figure 6. PRCCs for month-18 homeostatic margin under 500 Latin-hypercube parameter sets.

3.5 Fixed Parameter Stochastic Perturbations

Under 500 Euler-Maruyama trajectories with fixed structural parameters and illustrative diffusion amplitudes, 5 paths crossed by month 12, 495 crossed after month 12 but by month 18, and none remained uncrossed at the horizon. Median crossing time was 13.65 months (2.5^{th}-97.5^{th} percentiles 12.27-15.15). These distributions reflect the selected state noise specification and projection to [0,1]. Across RK45 and DOP853 combining at four maximum step settings each, terminal margin agreed to the displayed six decimal places (approximately -0.138742) and sampled crossing month remained 13.70.

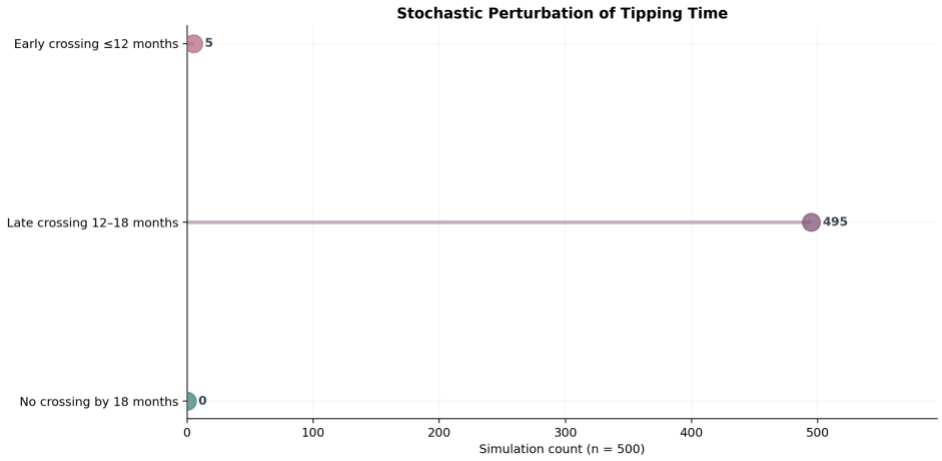


Figure 7. First passage classification of 500 fixed parameter stochastic paths, using the 12- and 18-month cutoffs.

3.6 Latent Quasi Equilibria and Local Stability

With observable functional and respiratory reserves held fixed, the two-dimensional M-H subsystem yielded quasi equilibria at each of three prespecified anchors. All calculated dominant Jacobian eigenvalue real parts were negative, supporting local asymptotic stability of this frozen subsystem at those anchors. These are not equilibria of the evolving ALS trajectory, and the calculation does not establish stability of the full four state system or prove a clinical bifurcation.

Table 5. Jacobian Eigenvalue

Anchor	ALSFRS-R / FVC	M*	H*	Dominant Re(λ)
Preserved	42 / 95%	0.661	0.484	-0.072
Compensated	38 / 80%	0.745	0.424	-0.090
Depleted	30 / 60%	0.829	0.340	-0.107

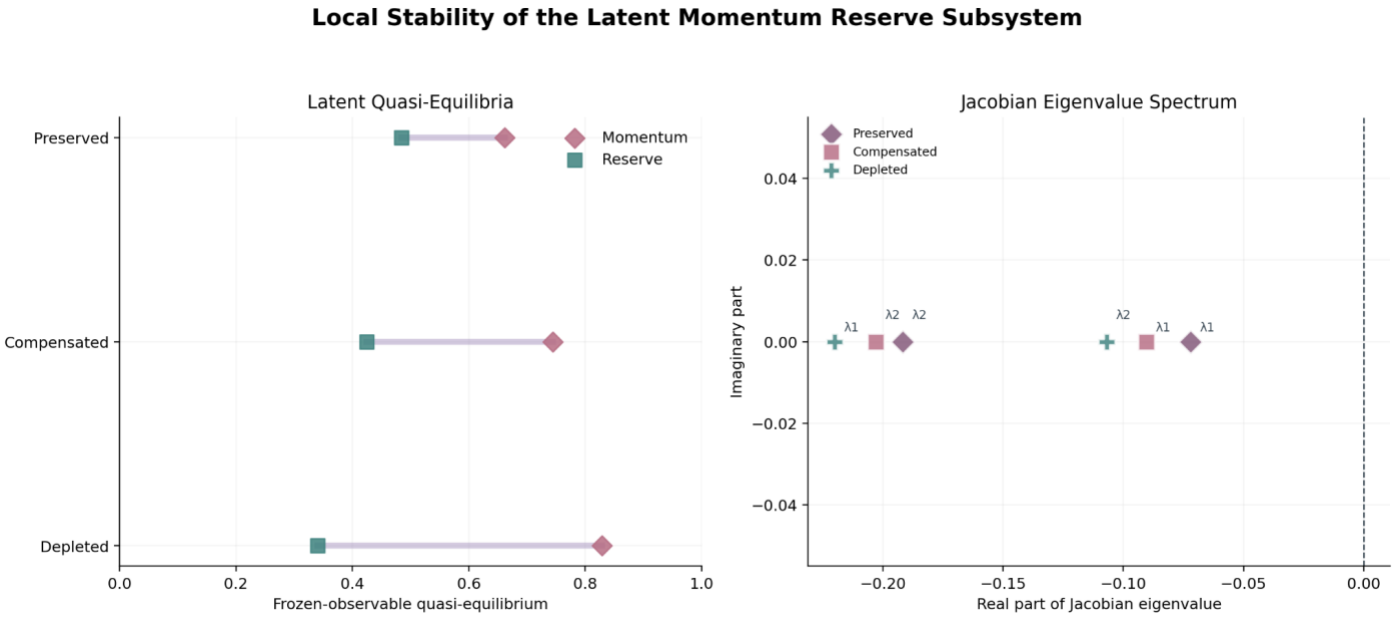


Figure 8. Latent quasi equilibria and eigenvalue spectra for three observable anchors.

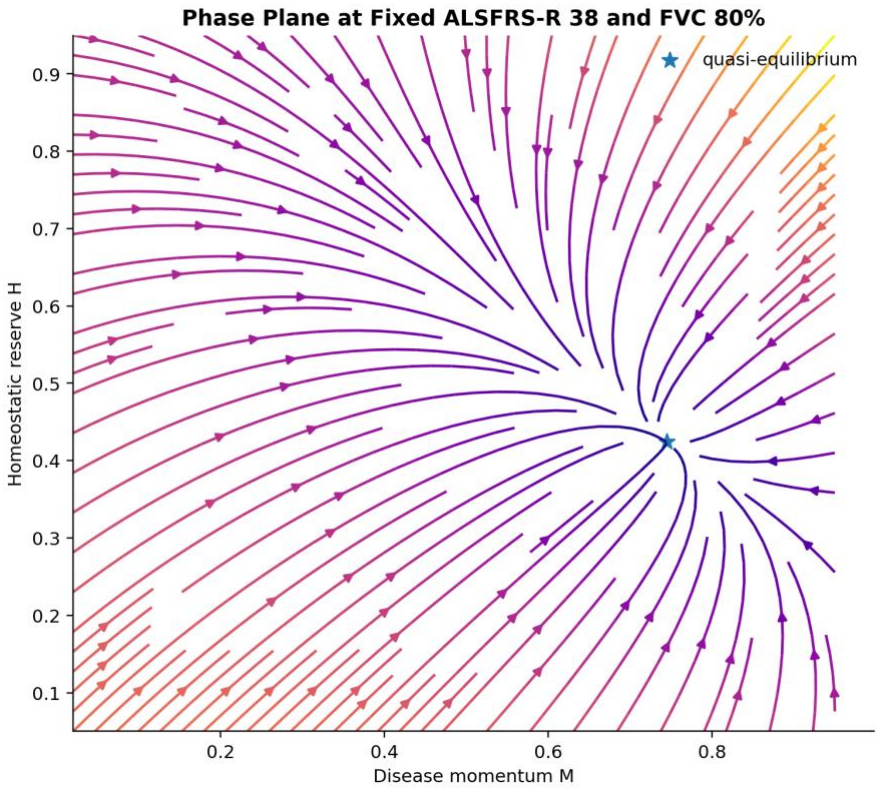


Figure 9. M-H phase portrait at fixed ALSFRS-R 38 and FVC 80%. Star denotes the conditional quasi equilibrium.

3.7 Synthetic Model Recovery

The internal benchmark compared four candidate specifications against monthly pseudo-observations produced from the reference model with stipulated Gaussian noise. Under the implemented standardized residual AIC* criterion, the producing structural model had AIC* -3.86, the shared inflection latent reserve model -1.19 (Δ 2.67), the independent linear model 11.53 (Δ 15.39), and the coupled exponential model 37.89 (Δ 41.74). Crucially, the benchmark counts θ_0 as a fitted structural parameter although θ_0 does not enter the simulated ALSFRS-R/FVC observations. Therefore, the structural AIC* and ΔAIC* are retained as archived and provisional code outputs.

Table 6. Internal Model Recovery Comparison

Candidate model	Archived AIC*	ΔAIC*	Assigned k
Full regime switching structural model	-3.86	0.00	4
Shared inflection latent reserve model	-1.19	2.67	6
Independent linear trajectory model	11.53	15.39	4
Coupled exponential reserve model	37.89	41.74	3

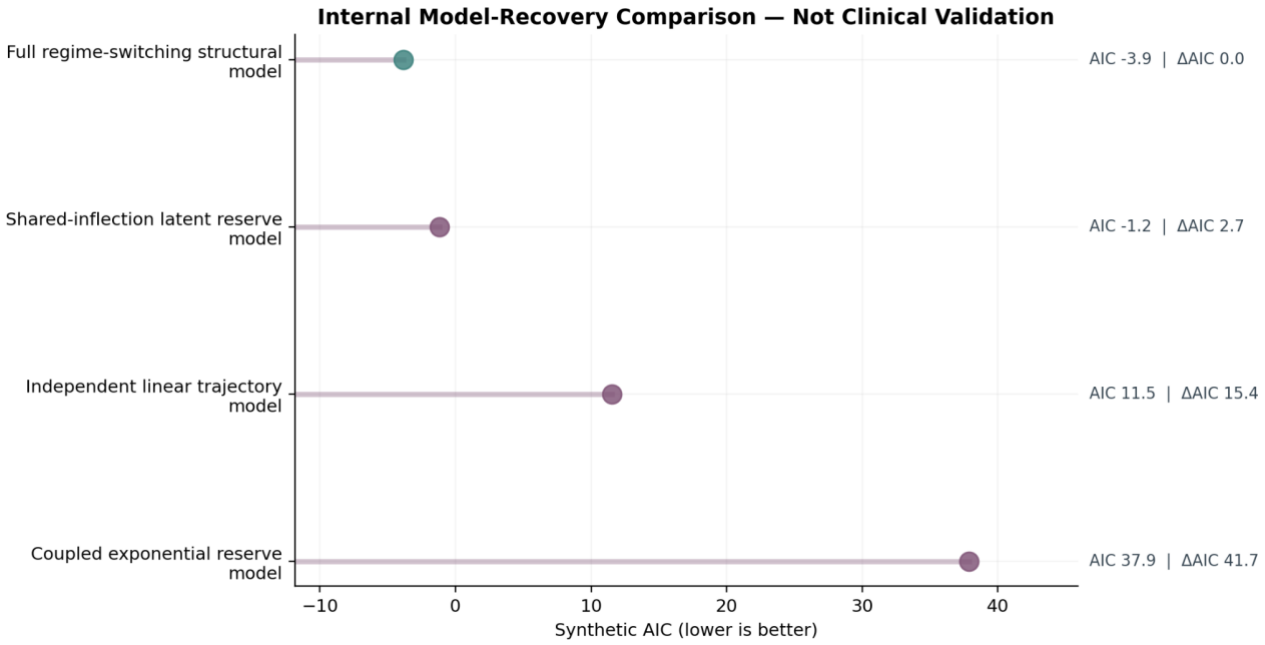


Figure 10. Archived synthetic standardized residual AIC* model recovery comparison.

3.8 Exploratory Three-Month Surface

The three-month decision layer propagated the same structural ODEs over baseline ALSFRS-R 30-44 and FVC 50-100% predicted, keeping latent initial conditions fixed. A logistic transformation of $\Psi(3)$ produced a theoretical score $s = \frac{1}{[1+\exp(10\Psi(3))]}$, with an illustrative escalated review label assigned when $s > 0.20$. Higher scores occurred for lower initial functional and respiratory values. The implemented procedure is a deterministic threshold comparison with an arbitrary monitoring cost value of 0.20. It cannot determine clinical follow up frequency or treatment decisions.

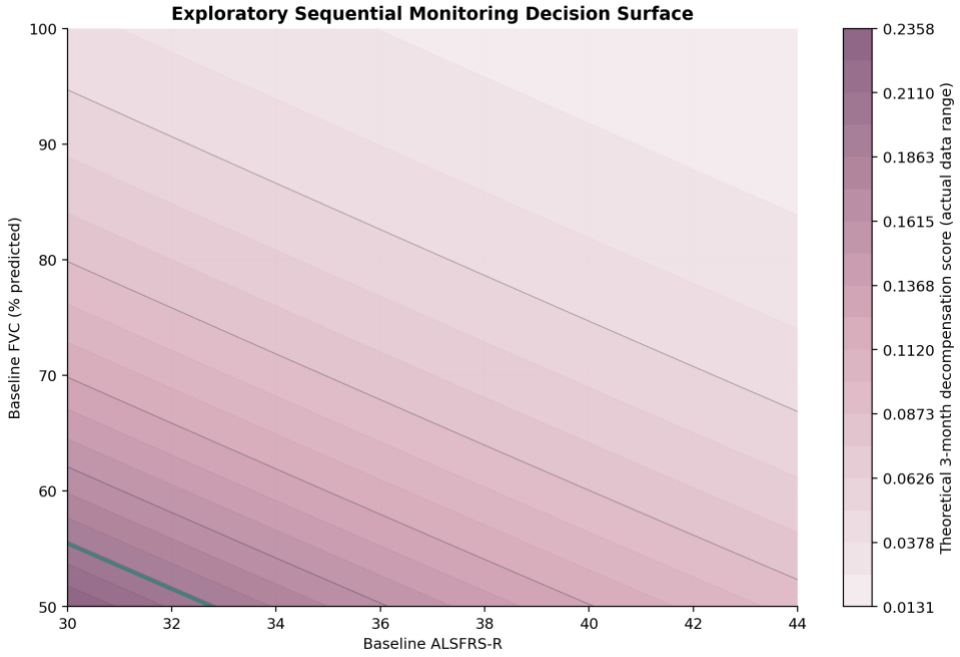


Figure 11. Theoretical three-month monitoring score from the model margin. Highlighted contour reflects the illustrative score threshold.

4. Discussion

The reference trajectory, oriented to published cohort average ALSFRS-R and vital capacity slopes, crossed the model margin $\Psi = 0$ at 13.70 months. Across the baseline grid, the month-12 margin varied with both initial ALSFRS-R and FVC, and structural stress tests shifted crossing times from 6.75 to 13.70 months, with no crossing observed by 18 months in the slow progression scenario. The main contribution is testable mathematical model: observable functional and respiratory trajectories jointly influence a state dependent margin through latent momentum M and homeostatic reserve H. The month-12 surface separated 323 of 442 sampled initial condition combinations below $\Psi = 0$, 104 in the interval $0 \leq \Psi < 0.10$, and 15 at $\Psi \geq 0.10$. These proportions describe the prespecified rectangular grid. The discrete continuation yielded a decreasing minimum baseline FVC for a nonnegative month-12 margin as baseline ALSFRS-R increased (99% at ALSFRS-R 30, 83% at 38, 80% at 40, and 73% at 44). The scenarios illustrate that distinct imposed conditions can reach an adverse model margin through different combinations of functional and respiratory decline. Respiratory vulnerability produced month-12 ALSFRS-R 22.53 and FVC 39.63%, whereas faster functional progression produced 18.16 and 51.58%, respectively. The corresponding crossing times were 9.65 and 10.00 months. Such contrasts support the utility of testing joint trajectories in silico. Similarly, the reported 12.95-fold reference hazard index change is a relative change in an imposed mathematical index.

At three frozen functional respiratory anchors, the latent M-H subsystem admitted quasi equilibria with negative dominant real Jacobian eigenvalues (approximately -0.072, -0.090, and -0.107). This supports local numerical stability of the latent subsystem while F and R constant and the model's monotonic F and R equations further limit its ability to represent transient improvements, measurement variability, and intervention related changes in observed trajectories. The 500-draw Latin-hypercube analysis gave a median terminal margin of -0.1507, with 2.5^{th}-97.5^{th} simulation percentiles of -0.3094 to -0.0016, 97.8% of sampled parameter configurations crossed $\Psi = 0$ by month 18. The strongest partial rank correlations included the boundary coefficients θ_M (-0.888) and θ_0 (-0.856), followed by k_R (-0.855), k_{MH} (+0.852), and k_F (-0.813). The importance of θ_M and θ_0 is consequential. The outcome is partly determined by the same assumed coefficients that define the boundary. PRCC magnitudes and signs therefore describe associations within the selected parameter ranges and model structure. Under 500 fixed parameter Euler-Maruyama perturbations, five paths crossed by month 12 and 495 between months 12 and 18, with median crossing at 13.65 months. These quantiles reflect the stipulated diffusion amplitudes and numerical projection. Agreement between RK45 and DOP853 across tested maximum steps supports numerical consistency for the reference calculation. In particular, near universal crossing under the chosen settings should not be extrapolated to an ALS cohort.

The synthetic model recovery exercise ranked the generating structural specification ahead of three simpler candidate curves under the archived standardized-residual AIC* calculation (-3.86 versus -1.19, 11.53, and 37.89). Because the pseudo-observations were produced from the reference system, this result primarily checks whether the fitting procedure can recover its own generating form. The three-month monitoring surface applies a logistic transformation to the computed terminal margin and assigns an illustrative escalated-review label at $s > 0.20$. Accordingly, the surface can motivate future design of monitoring studies but should not guide NIV initiation, gastrostomy, treatment choice, or surveillance intervals.

5. Limitations

This study is a literature-parameterized mathematical proof of concept rather than an analysis of patient level ALS trajectories. No individual PRO-ACT records or other registry observations were downloaded, fitted, or used for outcome validation. Published ALSFRS-R and vital capacity decrease rates were used to orient the early reference trajectory, but the remaining structural coefficients, initial latent states, margin weights, regime cutoffs, hazard-index coefficients, diffusion amplitudes, and monitoring threshold were assumed. The latent variables disease momentum M and homeostatic reserve H are mathematical constructs without established direct clinical measurements. Their identifiability from repeated ALSFRS-R and FVC/SVC observations has not been showed.

The proposed preserved, compensated, transitional, and decompensated regimes are defined by prespecified thresholds on $\Psi(t)$. They are not validated ALS stages and do not constitute an estimated Markov regime switching process. The zero crossing of Ψ is a model first passage event, not a biological tipping point, bifurcation, or irreversible clinical transition. The 442-cell baseline surface represents a deliberately selected rectangular grid, not a sampled patient population. Its cell proportions and stepped continuation boundary depend on grid resolution, assumed latent initial states, coefficient choices, and the 12-month evaluation horizon. The scenario crossing times (6.75-13.70 months among crossing profiles), the 97.8% crossing fraction in 500 Latin-hypercube draws, and the 12.27-15.15-month stochastic percentile interval are conditional simulation outputs, not incidence rates, individual prognosis, or empirically estimated confidence intervals.

The stochastic Euler-Maruyama experiment uses illustrative state diffusion rather than diffusion inferred from longitudinal patient variability. Agreement across RK45 and DOP853 solver configurations supports numerical repeatability for the reference equations but does not address structural misspecification. The propagated hazard multiplier is a theoretical index without an estimated event baseline clinical endpoint. The three-month monitoring surface is an illustrative logistic score and assumed action rule, not a validated event probability, or recommendation for treatment or respiratory intervention. The model currently excludes patient specific covariates, treatment effects, competing events, informative observation times, missing data, and the heterogeneity of clinical trial recruitment.

6. Conclusions

We analyzed and computationally examined a four-state nonlinear model coupling ALSFRS-R-derived functional reserve and vital capacity derived respiratory reserve with latent disease momentum and homeostatic reserve. Under literature reference rates and provisional coefficients, the model produced a reproducible homeostatic margin, a reference first passage at 13.70 months, and baseline dependent 12-month compensation surfaces. Scenario analysis, observable stability calculations, parameter sensitivity, stochastic perturbations, and numerical solver checks characterized the behavior and uncertainty of this specified mathematical system.

The central contribution is an explicit, falsifiable hypothesis such as joint functional and respiratory trajectories, considered with a latent state representation, may be investigated through a dynamic compensation margin and first passage analysis. This simulation is complementary to established ALS assessment and prognostic approaches. Its threshold defined regimes should not be described as empirically calibrated Markov switching, and the synthetic model recovery and exploratory monitoring results should remain separate from claims of clinical validation.

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